Dataformats: What should digis/stubs/clusters look like?

Stefano Mersi
Tracker DPG workshop
2026-10-16

Introduction

This talk is meant as an **introduction to the following discussion**.

Content (rather simple):

- Module local coordinates: how they are defined
- How the chips are arranged in the modules
- How hits are seen and transmitted by the front-end chips

The discussion follows:

- What should digits look like?
- Any related business: let's iron out the wrinkles!

Local axes definition

Module local axes are fixed for each module type

- Inner Tracker
  - 12 modules, 1 rule based on ROC0 rows and columns
  - Local axes presented at Inner Tracker Modules Meeting 2025-09-25
- Outer Tracker
  - 2 modules, 1 rule based on “upper/right” front-end numbering
  - Local axes fixed a long ago by module design

Local axes definition

Module local axes are fixed for each module type

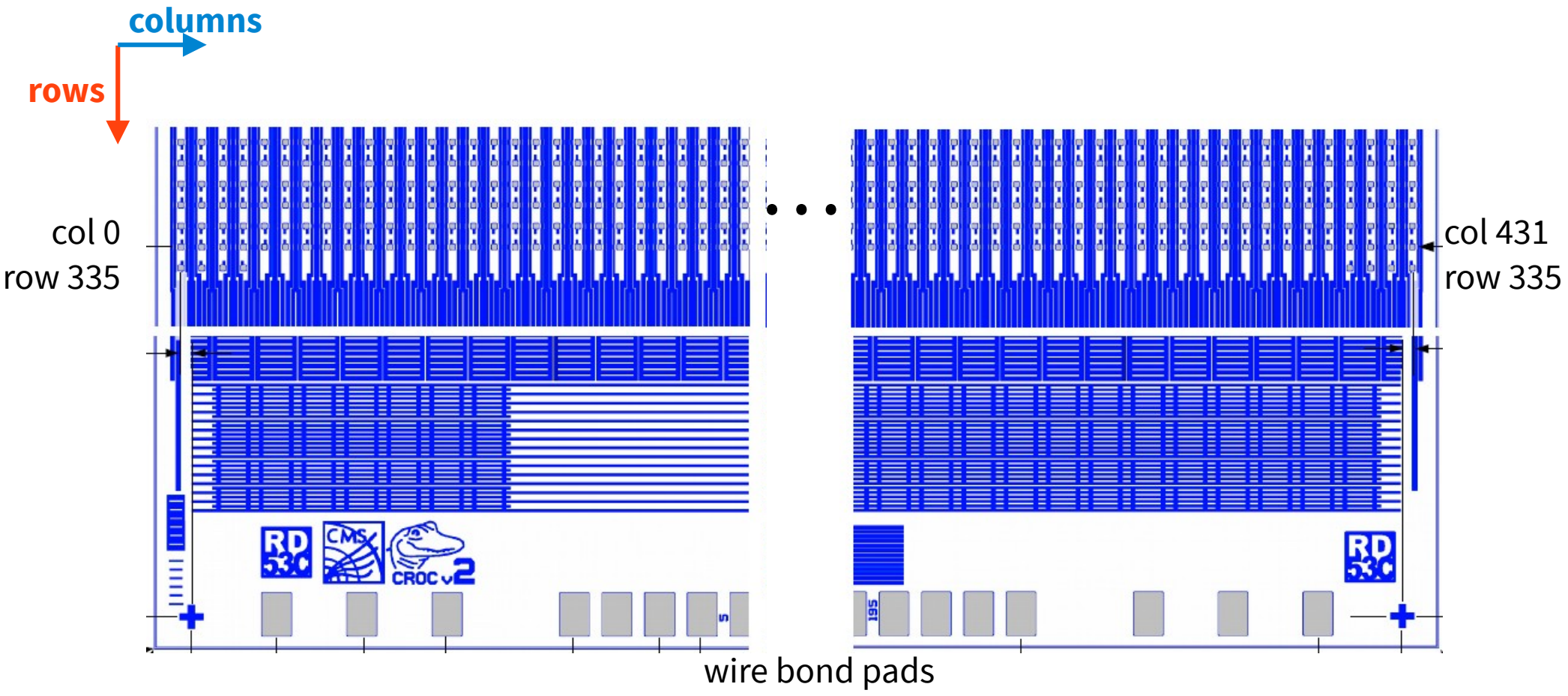
- Inner Tracker
  - 12 modules, 1 rule based on ROC0 rows and columns
  - Local axes presented at Inner Tracker Modules Meeting 2025-09-25
- Outer Tracker
  - 2 modules, 1 rule based on “upper/right” front-end numbering
  - Local axes fixed a long ago by module design

Inner Tracker: 12 module types

- One HDI design per family, configured as one of 12 module types (tkLayout sub-type 1–12)
- The type fixes chip placement, chip orientation and e-link routing
- Refer to modules by type, not by layer/ring:
  - TEPX R2 module at 12 o'clock uses the R3 wiring (type 10)
  - type 7 (TFPX R4) is wired as type 6 but installed in the opposite direction

Type	Location	Chips
1	TBPX L1	2
2	TBPX L2	2
3	TBPX L3	4
4	TBPX L4	4
5	TFPX R1–R2	2
6	TFPX R3	4
7	TFPX R4 (reversed)	4
8	TEPX R1	4
9	TEPX R2, mod. 2–7	4
10	TEPX R2 mod. 1, R3	4
11	TEPX R4	4
12	TEPX R5	4

CROC pixel numbering



Source: Gervasio Gomez's presentation at the Inner Tracker Modules meeting

Axes definition IT modules — TBPX

All module types

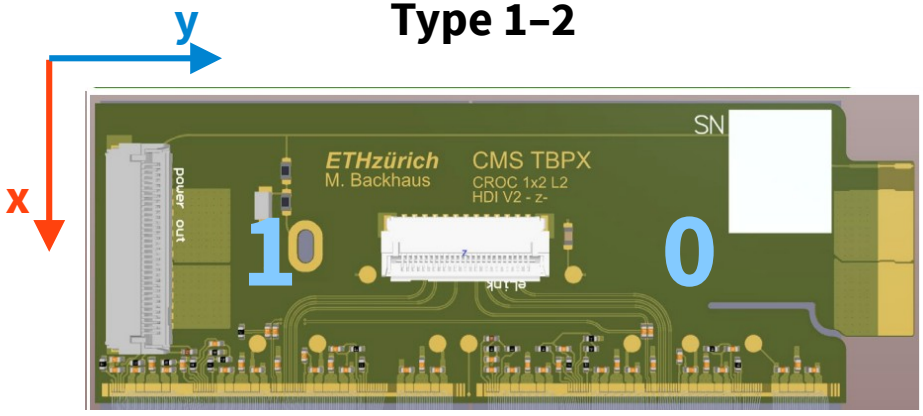
Chip 0 and 1

- x growing with rows
- y growing with columns

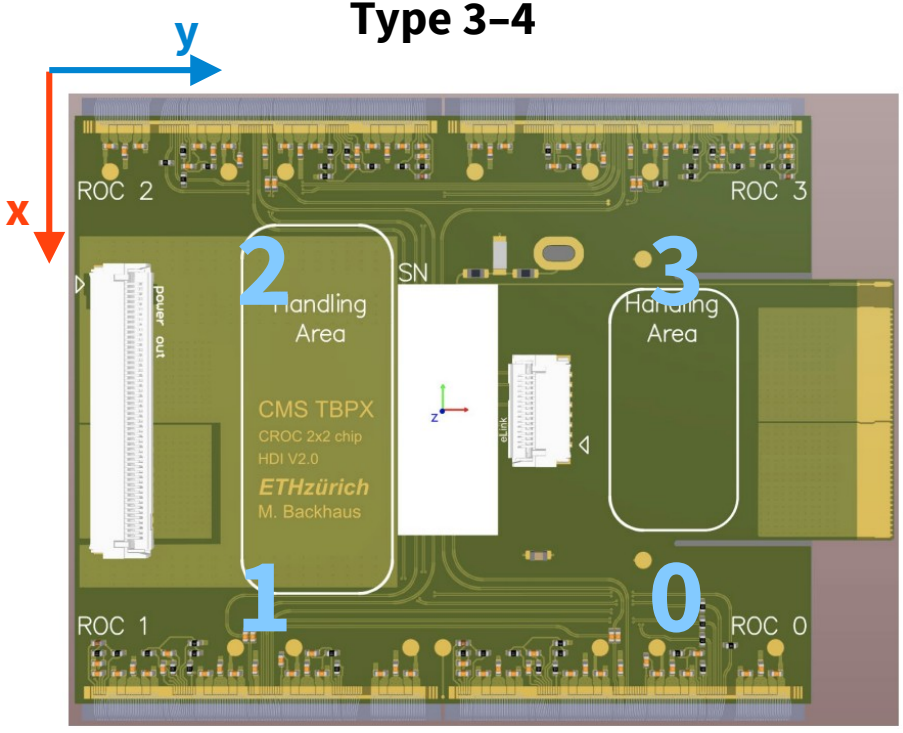
Other chips (if present) will adapt accordingly

CROC numbering and
module type definition from: HDI document on SharePoint

Type 1–2



Type 3–4



Axes definition IT modules — TBPX

All module types

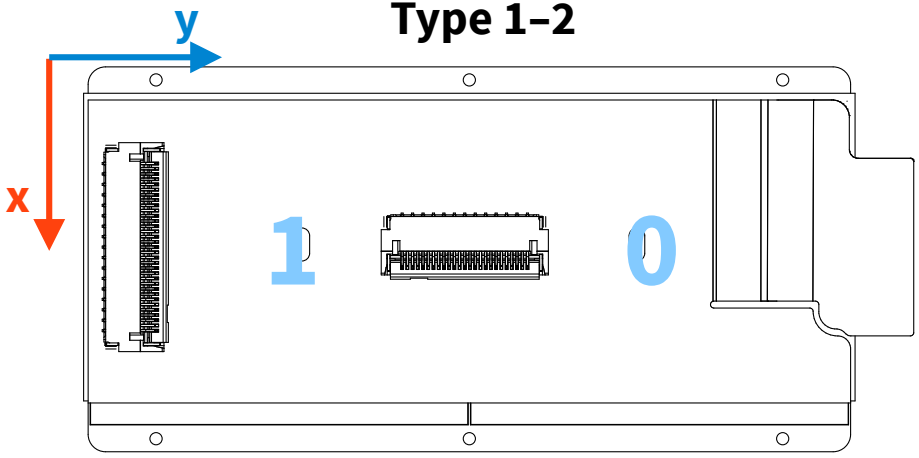
Chip 0 and 1

- x growing with rows
- y growing with columns

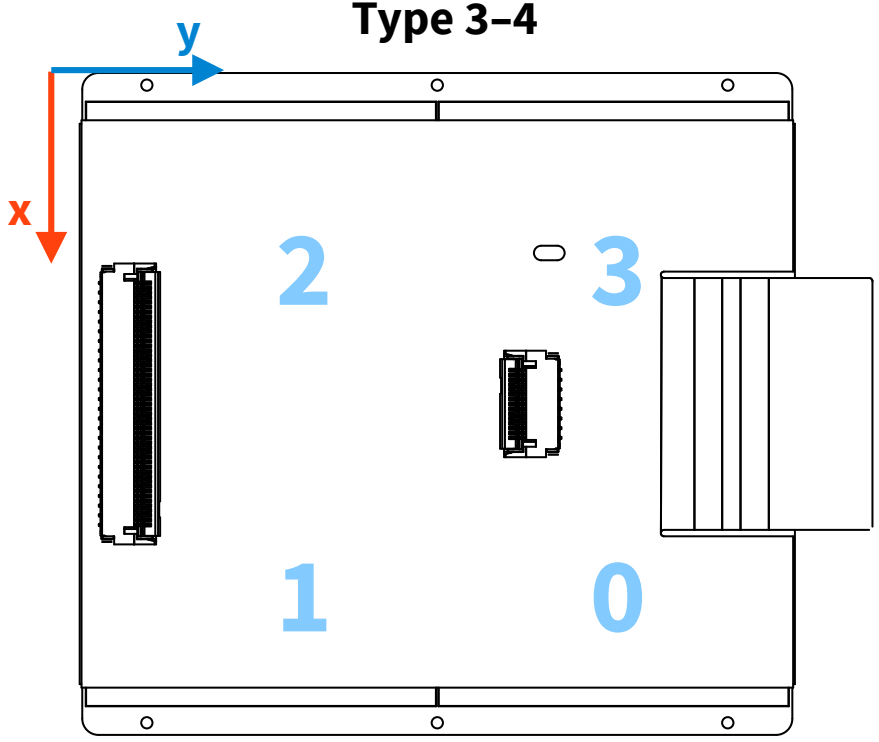
Other chips (if present) will adapt accordingly

CROC numbering and
module type definition from: HDI document on SharePoint

Type 1–2



Type 3–4



Axes definition IT modules — TFPX

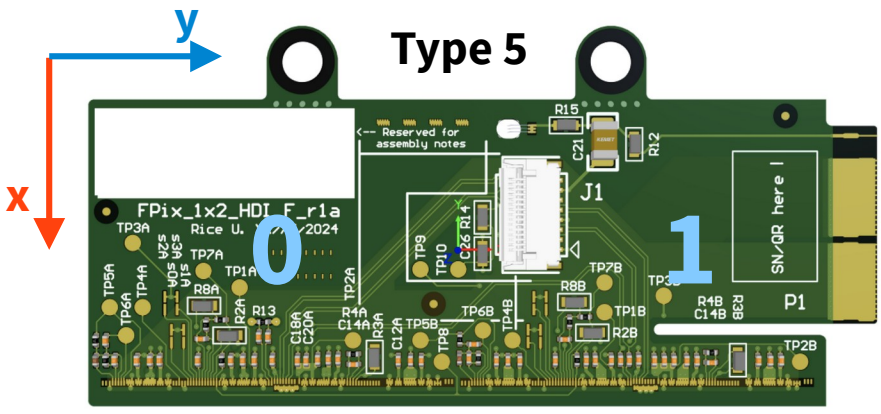
All module types

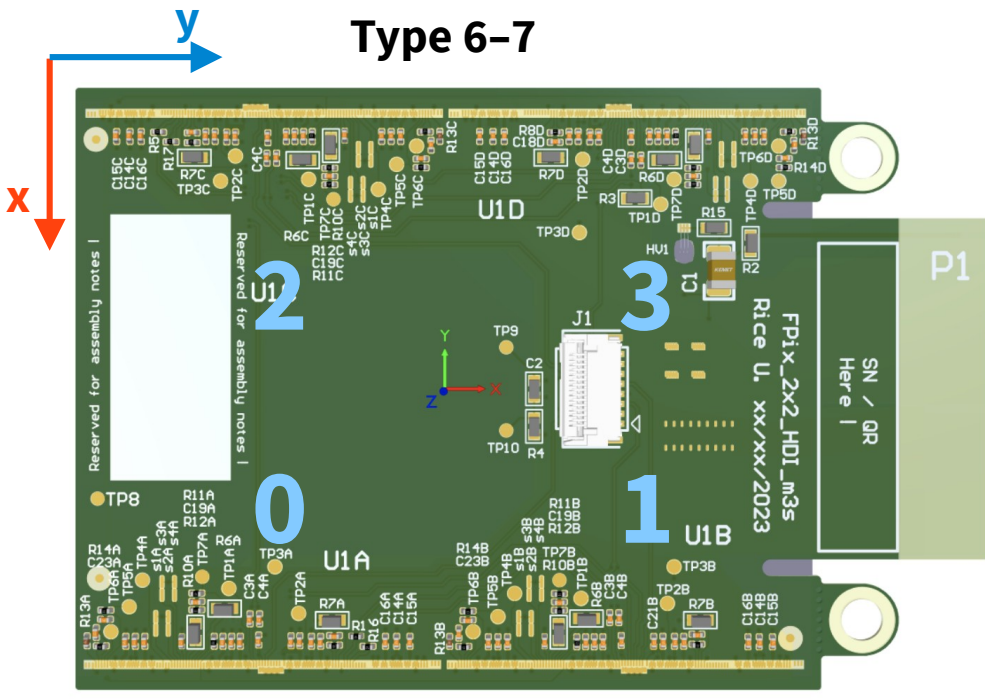
Chip 0 and 1

- x growing with rows
- y growing with columns

Other chips (if present) will adapt accordingly

CROC numbering and
module type definition from: HDI document on SharePoint





Axes definition IT modules — TEPX

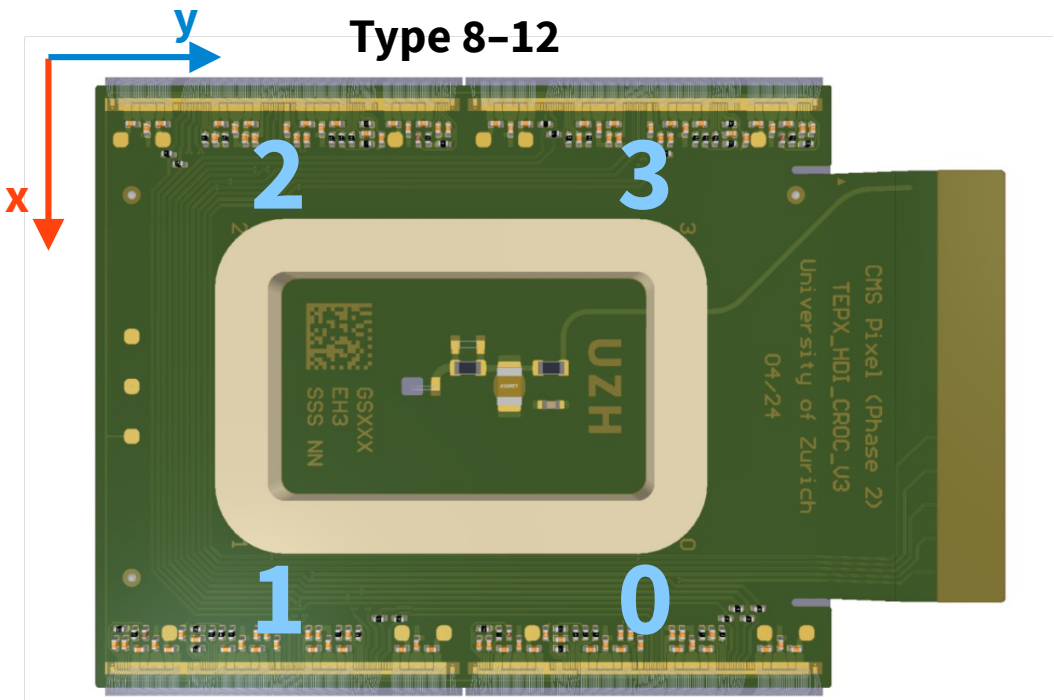
All module types

Chip 0 and 1

- x growing with rows
- y growing with columns

Other chips (if present) will adapt accordingly

CROC numbering and
module type definition from: HDI document on SharePoint



Axes definition IT modules — TEPX

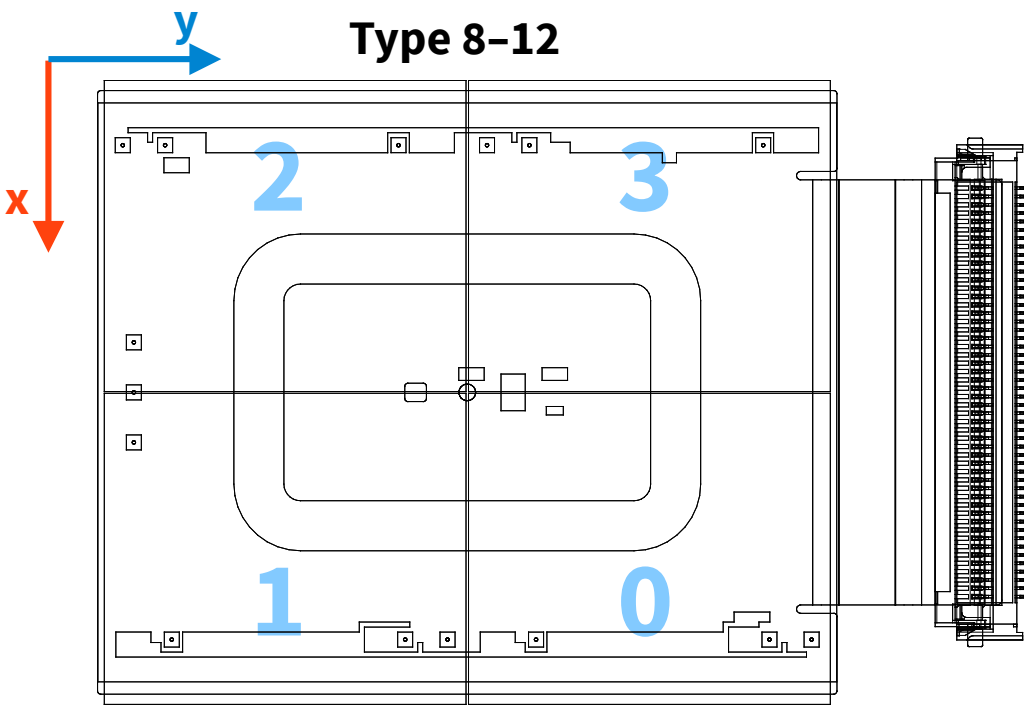
All module types

Chip 0 and 1

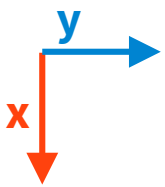
- x growing with rows
- y growing with columns

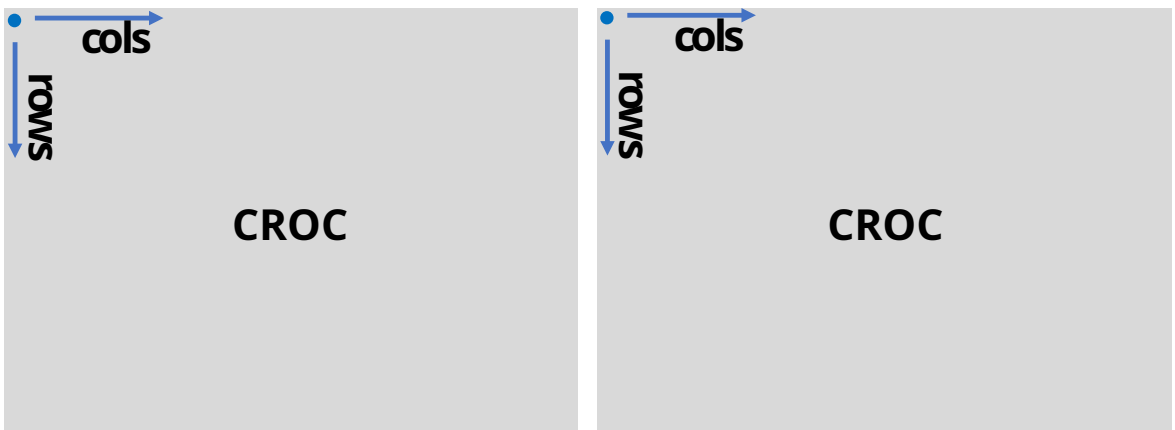
Other chips (if present) will adapt accordingly

CROC numbering and
module type definition from: HDI document on SharePoint



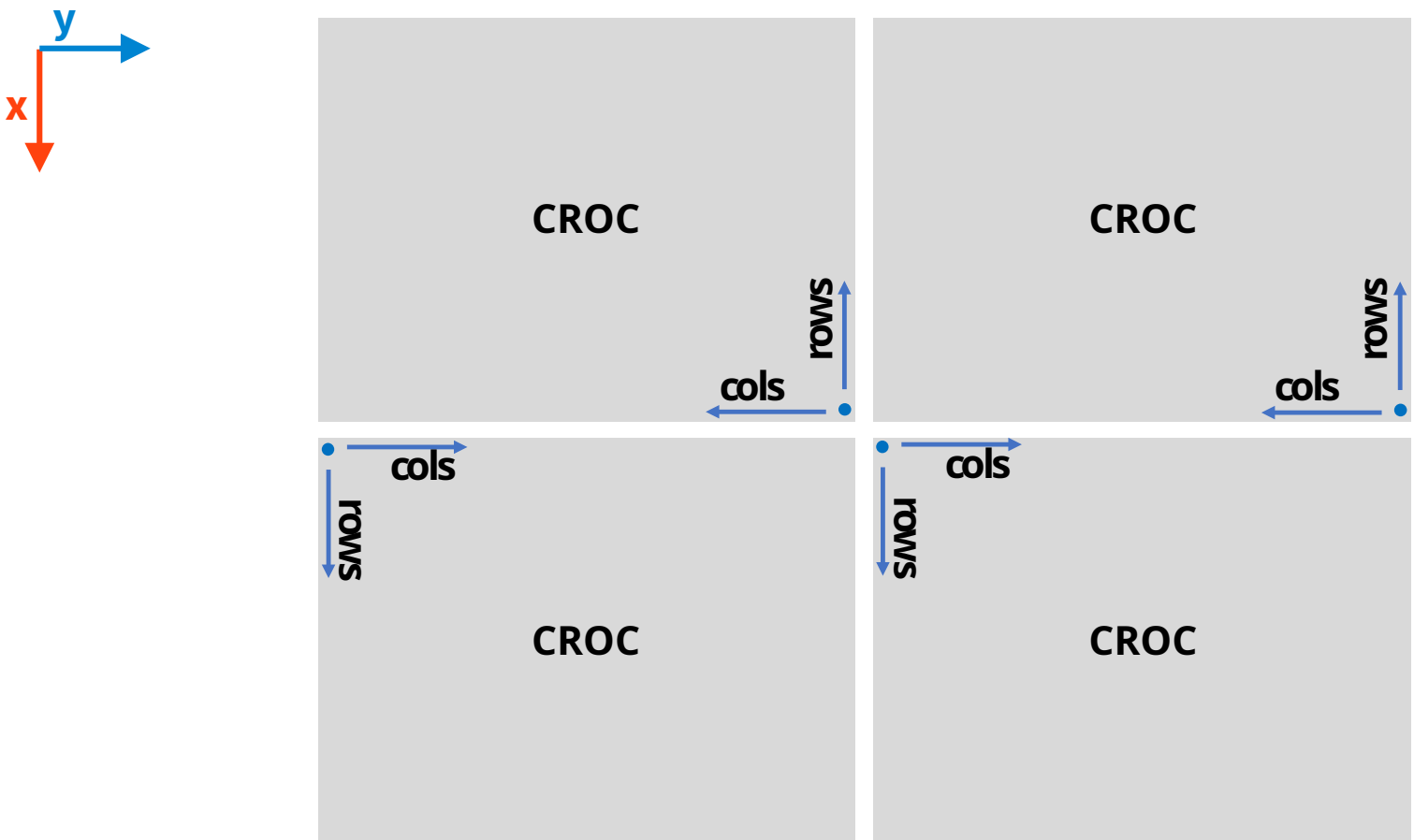
Dual module internal chip orientation





Source: Gervasio Gomez's presentation at the Inner Tracker Modules meeting

Quad module internal chip orientation



Source: Gervasio Gomez's presentation at the Inner Tracker Modules meeting

IT: chip channel vs sensor pixel

- The chip addresses channels on a $50 \times 50\ \mu m^2$ bump grid;
  sensors have $25 \times 100\ \mu m^2$ pixels
- Fixed bump-bonding pattern (CROC manual, R0C0 option),
  same for planar and 3D:
  - $r_{sensor} = 2 \times r_{chip} + (c_{chip} \bmod 2)$
  - $c_{sensor} = \lfloor c_{chip} / 2 \rfloor$
- Physical pixel size is not uniform:
  “big pixels” at internal chip seams and borders
  - dual: 25×225 along the seam; quad: also 112×100, and 112×225 at the centre
  - this edge map must be in the CMSSW local geometry

IT: how the ToT is measured and reported

- ToT is internally measured (iToT) with a very large dynamic range
- The conversion depends on a **single linear** relation, with two sources of error:
  - The whole chip has a measurable response Q→iToT, but each pixel has a different one with a ±10% spread
  - The same charge may get different ToT depending on the time of arrival (or time of injection for the calibration)
- iToT is then sent out with a 4-bit resolution (0–14, with 15 being a reserved code).

  Two modes of operation
  - Single slope (S)
  - Dual slope (D)

	iToT	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38
ToT	**S**	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
	D	0	1	2	3	4	5	6	7	8	8	8	8	9	9	9	9	10	10	10	10	11	11	11	11	12	12	12	12	13	13	13	13	14	14	14	14	14	14	14

Data transmission details

- Chip-to-chip internal connections via HDI (data merging) allow to optimize the e-link bandwidth usage.
- Internal e-link arrangement (data merging and multi-lane) is handled by the firmware. Data will be shipped to HLT based on chip index (0–3), row/column and ToT.
- Nevertheless, the individual chip-to-chip and chip-to-lpGBT information is available in the cabling database, for e-link occupancy studies.

Type	Mode	N_{links}	Chip 0	Chip 1	Chip 2	Chip 3
1	Multi-lane	6	0,1,2	3,4,5	—	—
2	Default	2	0	1	—	—
3	Data merging	2	0	0	2	2
4	Data merging	1	0	0	0	0
5	Multi-lane	3	0,1	2	—	—
6	Data merging	2	0	0	2	2
7	Data merging	2	0	0	2	2
8	Default	4	0	1	2	3
9	Data merging	3	0	1	2	0
10	Data merging	2	1	1	2	2
11	Data merging	2	0	3	0	3
12	Data merging	1	2	2	2	2

Output e-link index on which each chip's data appear, per module type. Entries with several values (types 1, 5) are multi-lane chips that use more than one link. A merged chip shares the link of the neighbour that serialises its data (e.g. type 3: chip 1 merges into chip 0 on link 0, chip 3 into chip 2 on link 2). Link numbers are an integer relabelling and do not follow the electronics naming convention.

All details here

Local axes definition

Module local axes are fixed for each module type

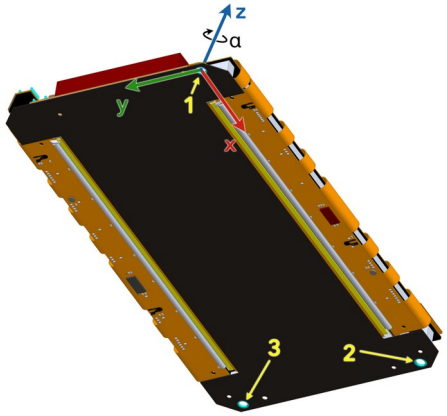
- Inner Tracker
  - 12 modules, 1 rule based on ROC0 rows and columns
  - Local axes presented at Inner Tracker Modules Meeting 2025-09-25
- Outer Tracker
  - 2 modules, 1 rule based on “upper/right” front-end numbering
  - Local axes fixed a long ago by module design

Local axes definition

Module local axes are fixed for each module type

- Inner Tracker
  - 12 modules, 1 rule based on ROC0 rows and columns
  - Local axes presented at Inner Tracker Modules Meeting 2025-09-25
- Outer Tracker
  - 2 modules, 1 rule based on “upper/right” front-end numbering
  - Local axes fixed a long ago by module design

Axes definition OT modules

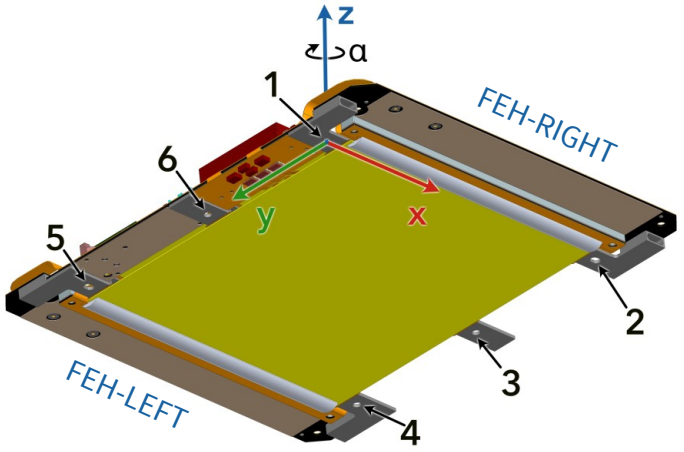


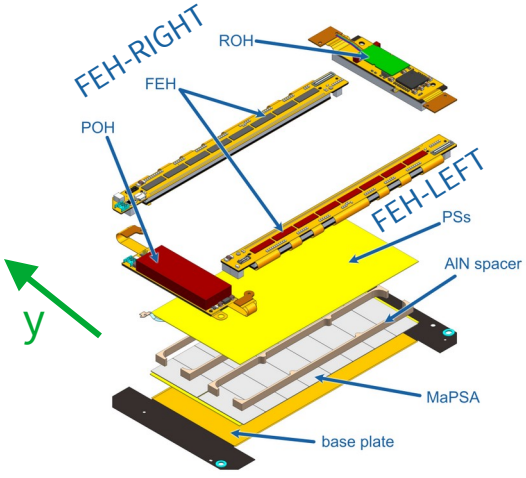
2S module

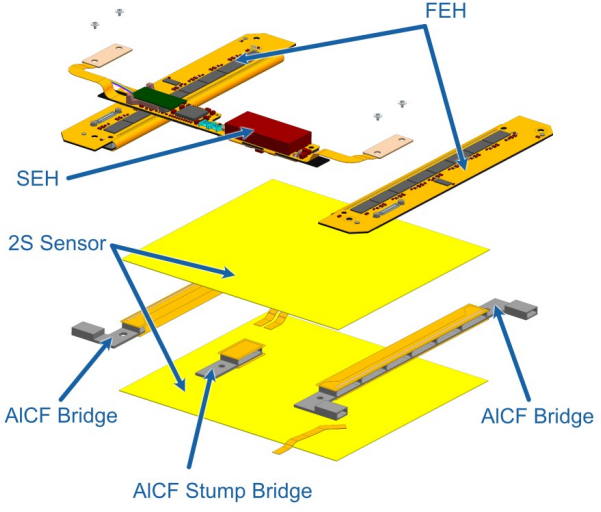
- x away from DC/DC
- y from right to left
- z from bottom to top

PS module

- x away from DC/DC
- y from left to right
- z from bottom to top







Comments on module local x axes

Well-established definition of axes, anchored to the DC/DC converter. But front-end hybrids are numbered 0=right and 1=left **as seen from the lpGBT**, which is on the same side of DC/DC in 2S and opposite in PS.

Warning:

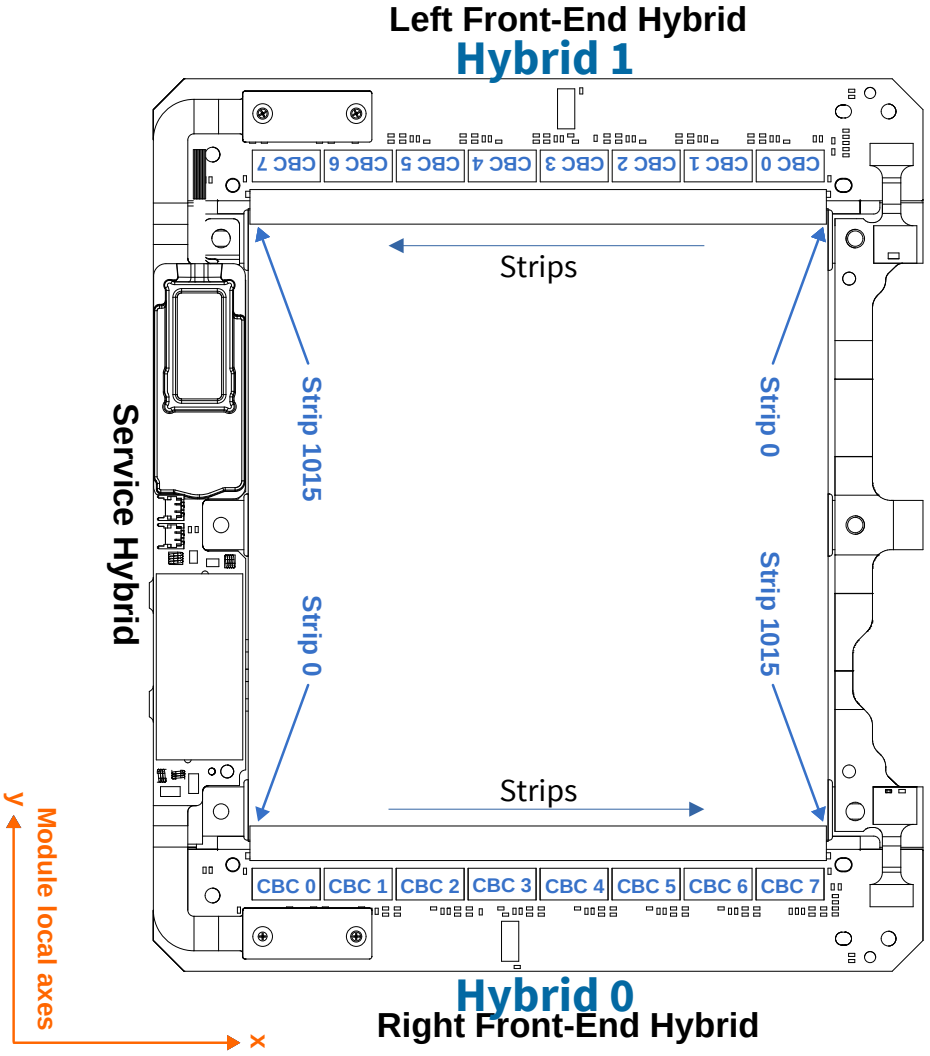
- In **2S modules**:
  - **x axis** goes from **low to high** strip indices in **hybrid 0** (right)
  - **x axis** goes from **high to low** strip indices in **hybrid 1** (left)
  - **y axis** goes from **hybrid 0** (right) to **hybrid 1** (left)
- In PS modules the
  - **x axis** goes from **high to low** strip indices in **hybrid 0** (right)
  - **x axis** goes from **low to high** strip indices in **hybrid 1** (left)
  - **y axis** goes from **hybrid 1** (left) to **hybrid 0** (right)

tkLayout does not export to CMSSW any detail of the module

Question: which step is responsible for the conversion of information based on sensor coordinates into CROC rows and columns?

2S chip arrangement

Many thanks to F. Ravera!



Raw data format

The back-end (firmware) will report the hits in terms of

- chip id
- sensor (upper/lower)
- strip number

Digi format

Same?

Converted into local "position"?

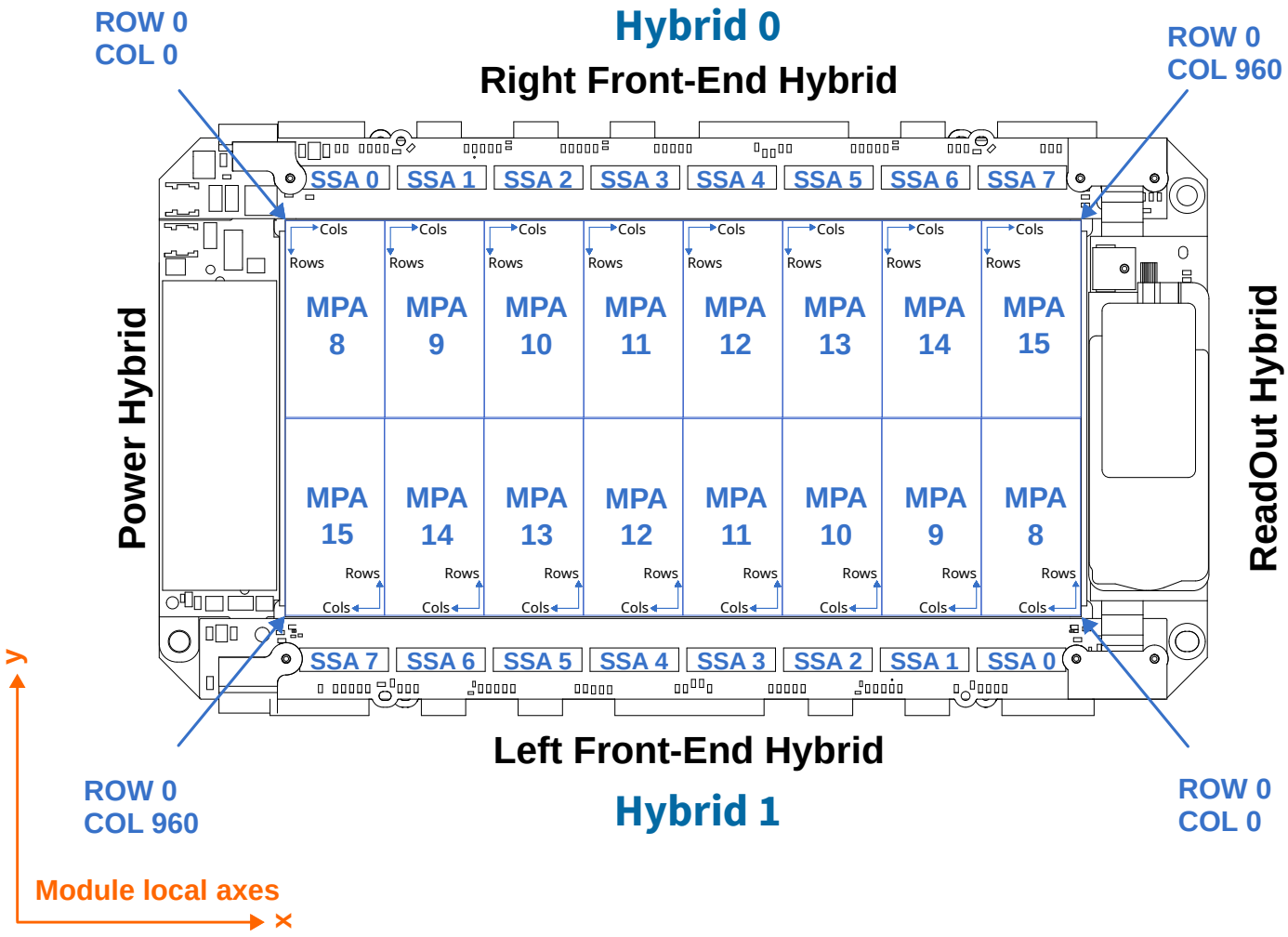
- **Upside**: more "logical" ?
- **Downside**: loses the link to the actual digital information

Notes on 2S data

- The CBC ships the unsparsified strip pattern; the CIC zero-suppresses it
- A run of consecutive hit strips on one sensor is sent once as a strip group: (sensor, address of the first strip, run length)
- Groups are per sensor: upper and lower sensors are sent separately
- Group length is a 3-bit field: at most 8 strips per group
- Longer groups, or groups split by a chip boundary, become several touching groups
- Re-joining them is offline clusterization
- The CIC calls these objects “clusters”: beware that they are not the same as offline “clusters”
- The stub path groups the same hits a second time, in the CBC, with a different algorithm: the two groupings need not coincide

PS chip arrangement

Many thanks to F. Ravera!



Raw data format

The back-end (firmware) will report the hits in terms of

- chip id
- column
- row

Digi format

Same?

Converted into local "position"?

- **Upside**: more "logical" ?
- **Downside**: loses the link to the actual digital information, but it still requires the knowledge of the specific channel arrangement (big pixels, etc)

Notes on PS data

- Sparsified already in the front-end (SSA, MPA); the CIC only aggregates and forwards
- Two cluster lists per event, one per sensor:
  - strip cluster: (first column, run length, HIP flag)
  - pixel cluster: (first column, run length, row)
- The list a hit arrives in is its local z: strip = top, macro-pixel = bottom
- Chip-level column address: 7 bits (0–119); the chip id gives the high part of the module-wide column
- SSA group length is 3 bits: at most 8 channels; longer runs become touching clusters
- MPA = ? (Ian, here, took a deep dive into this)
- Limited number of clusters per chip per event: rare truncation at high occupancy
- SSA “high charge” threshold is reported per group of strips. When adjacent groups have different “high charge” value, a decision must be taken when clusterizing.
- What is a useful “high charge” threshold? Simulation should guide calibration.

Conclusions

Facts:

- Module local axes are clearly defined
- Internal chip addressing is also well-defined

Pseudo-random selection of possible questions:

- What should digis represent?
- Can we afford 2×10^9 calibration constants in the pixels
  (not necessarily one byte each)
- How do we match the ToT with the most probable charge that generated it?
- Is it useful (and possible) to record individual chip→lpGBT in the offline cabling database?
- What is a useful “high charge” threshold for the SSA?
- …
- (I think you might have more)